

LLVM Release Process

EuroLLVM 2026 Status Update

Cullen Rhodes, Douglas Yung, Tobias Hieta

Introductions

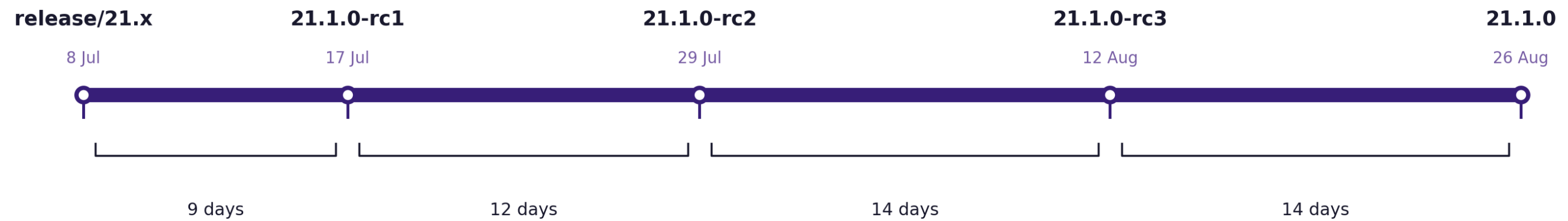
Recent Changes

- Two new Release Managers: **Cullen** and **Douglas**
- Stricter enforcement of acceptance criteria
- Automation updates
- Dropped split source packages

Current Process

- Major release every **6 months**
- Branch: 2nd Tuesday in **January** (even) / **July** (odd)
- Point releases every 2 weeks, typically through X.1.5 or X.1.6

LLVM 21 Release Timeline



20.x RC1: 16 days late · 21.x RC1: 6 days late · 22.x RC1: on time

RC Acceptance Criteria

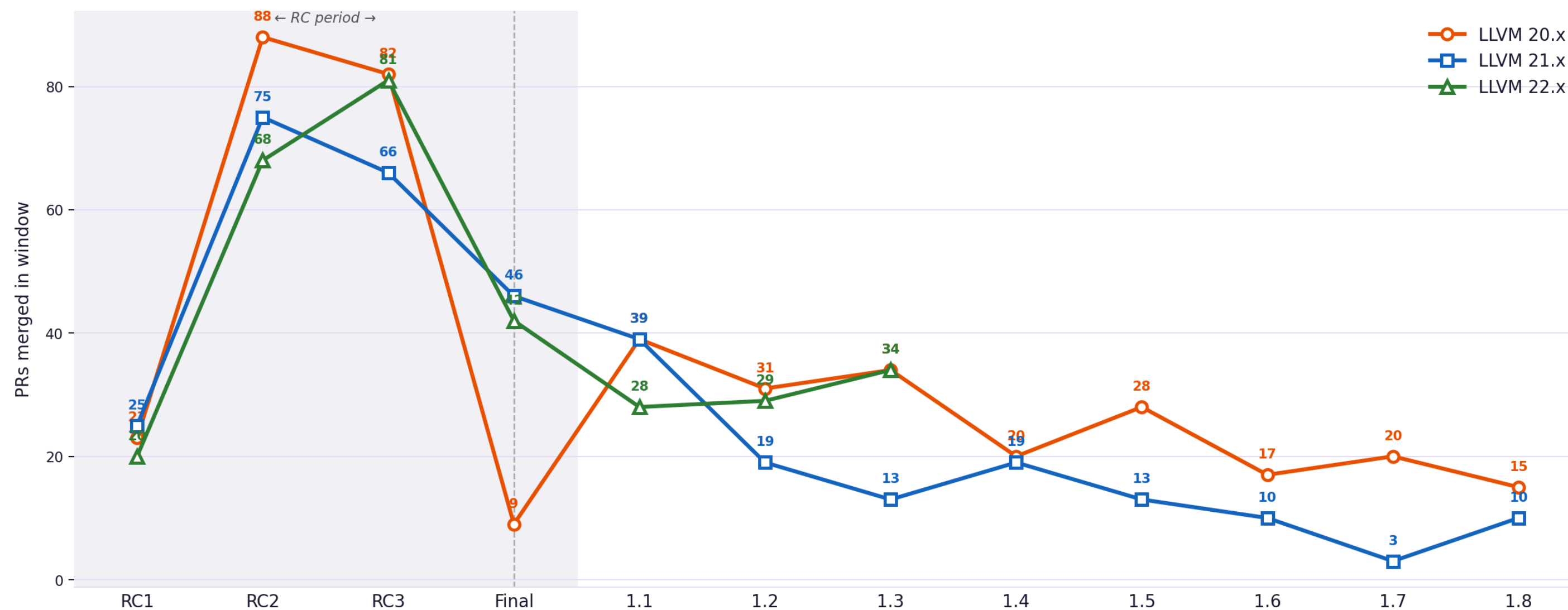
Phase	Accepted
RC1	Bug fixes, important optimizations, completion of features started before branch
RC2 / RC3	Bug fixes, very safe backend-specific improvements
Final (X.1.0)	Critical bugs and regressions only
Point releases	Bug fixes, critical performance, must maintain API+ABI compat

By the Numbers

Release	PRs Merged	PRs Rejected	Issues
20.x	406	53	125
21.x	343	68	122
22.x	309	76	89

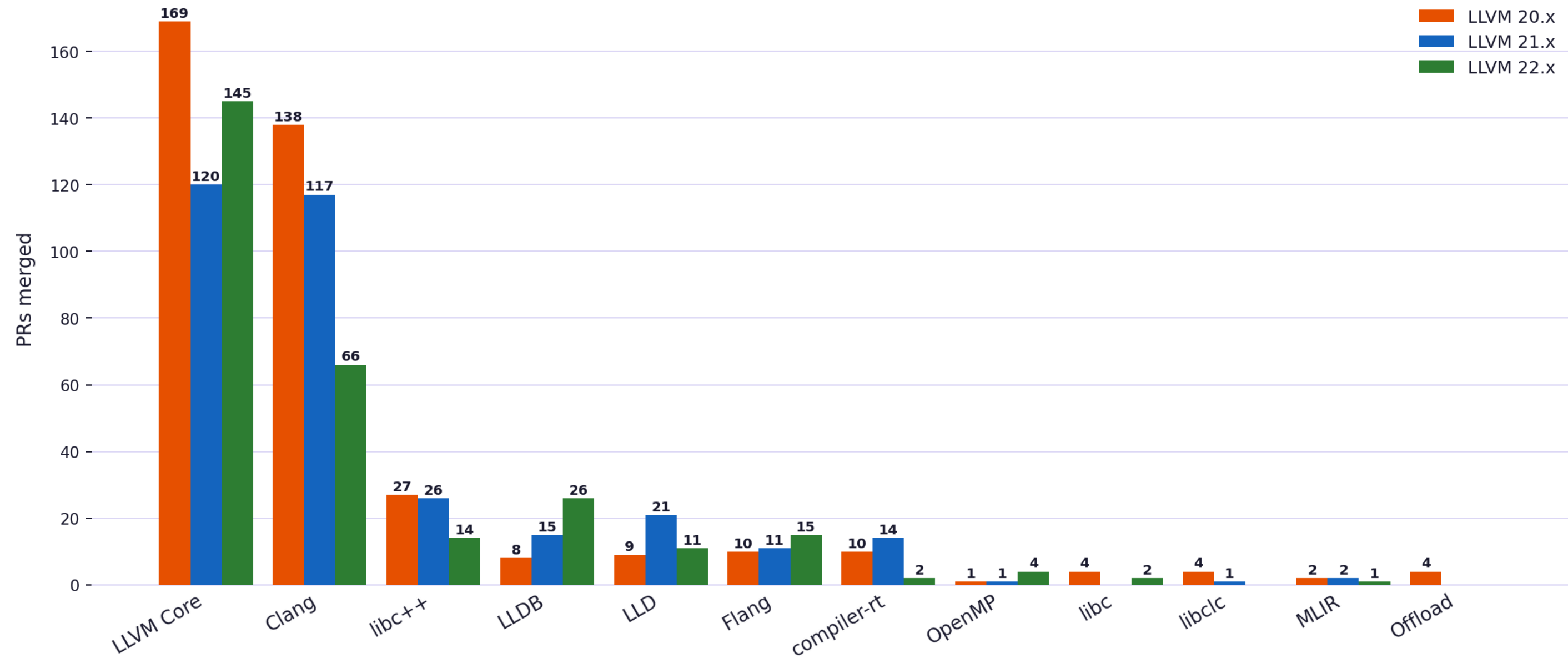
Rejection rate rising: **12%** → **17%** → **20%**

PRs Merged by Phase

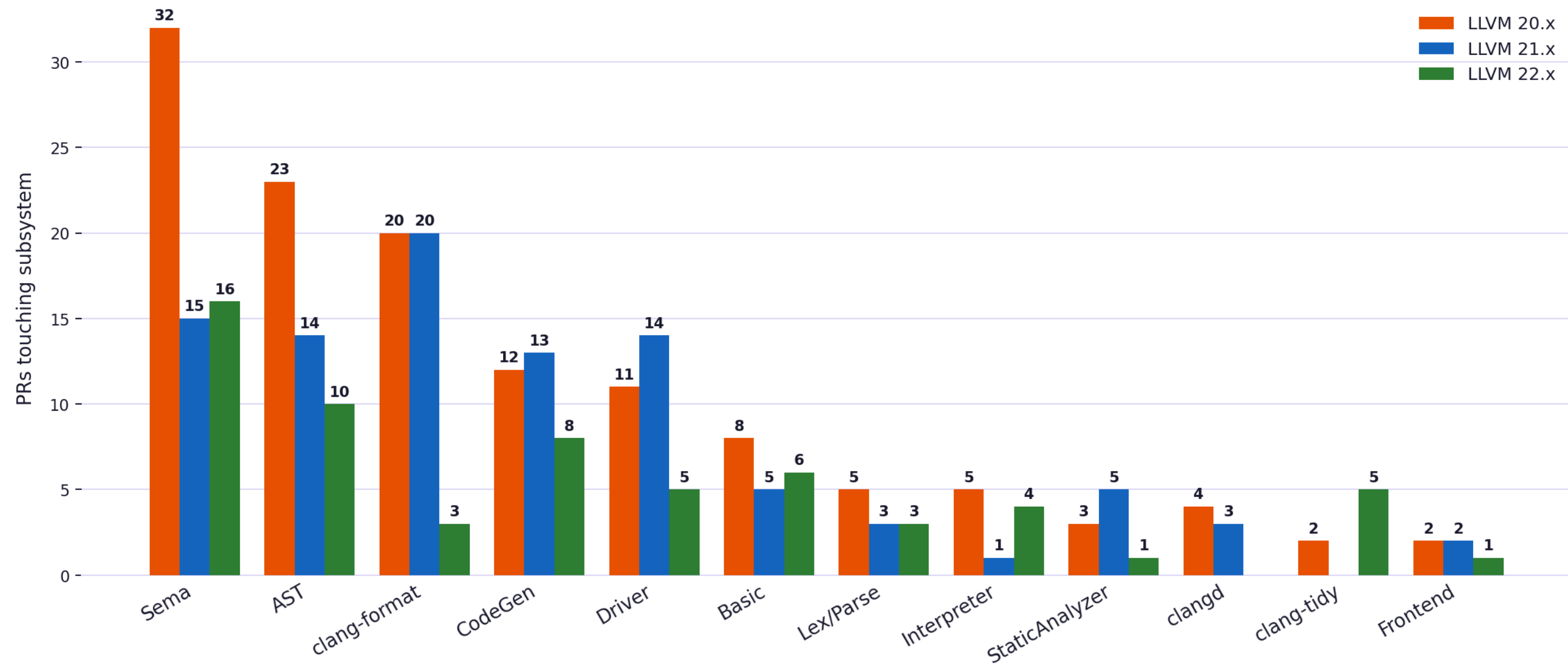


RC1→RC2 always **3× busier** than Branch→RC1 · RC3→Final: **9 PRs** (20.x) vs **42–46** (21.x / 22.x)

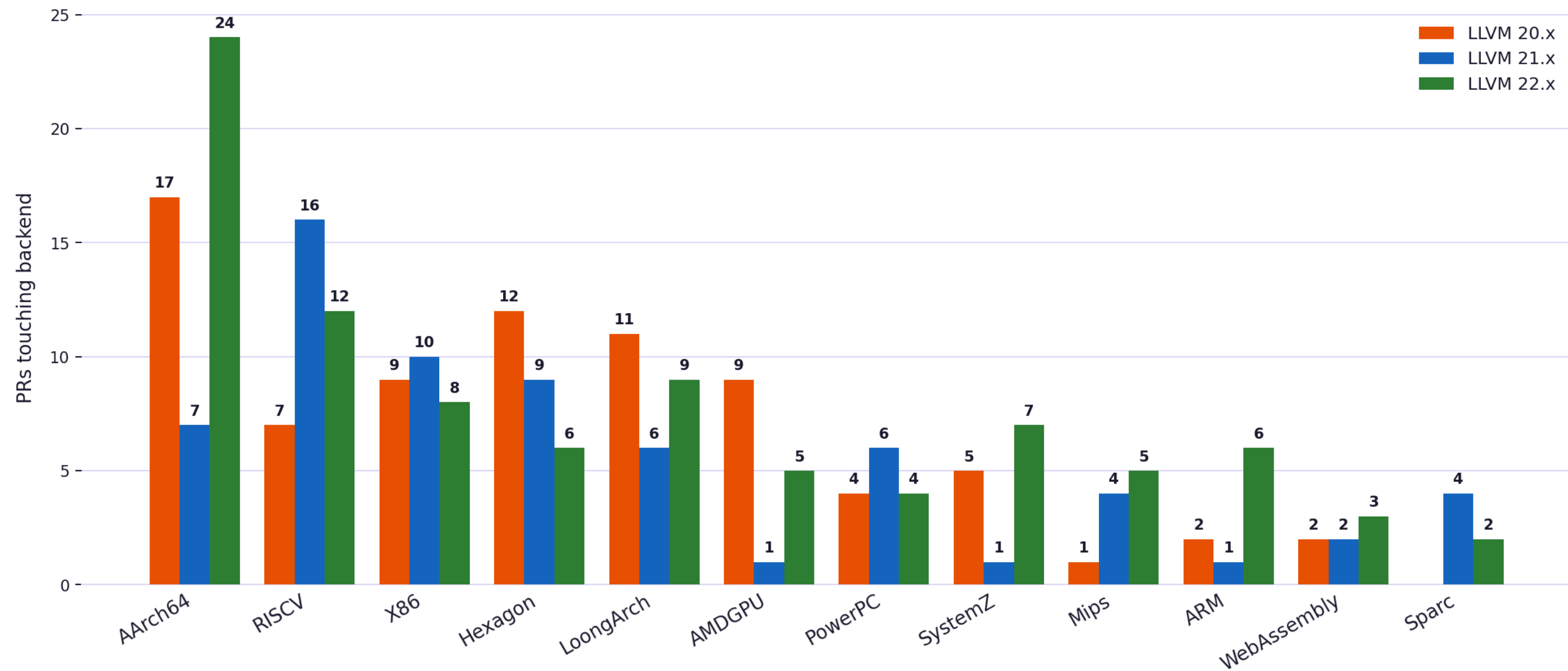
PRs by Subproject



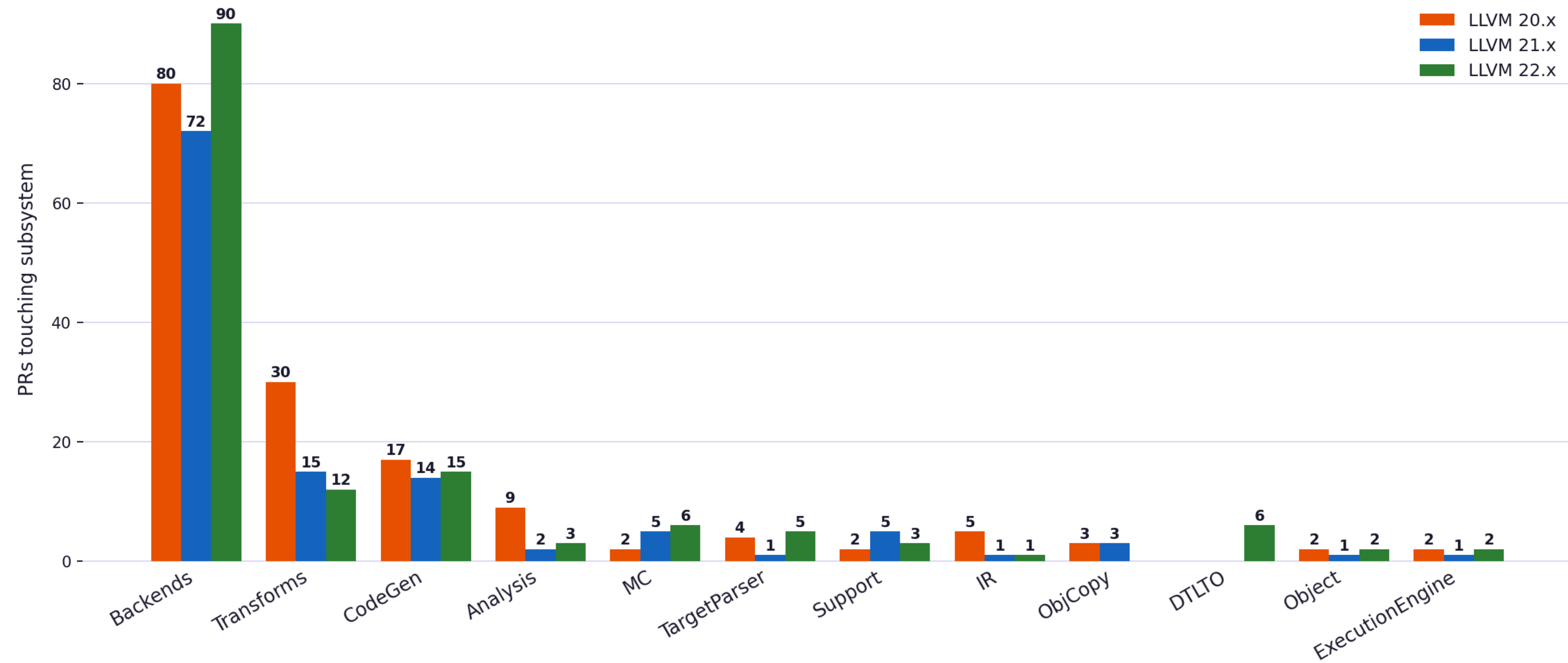
Clang: Subsystems



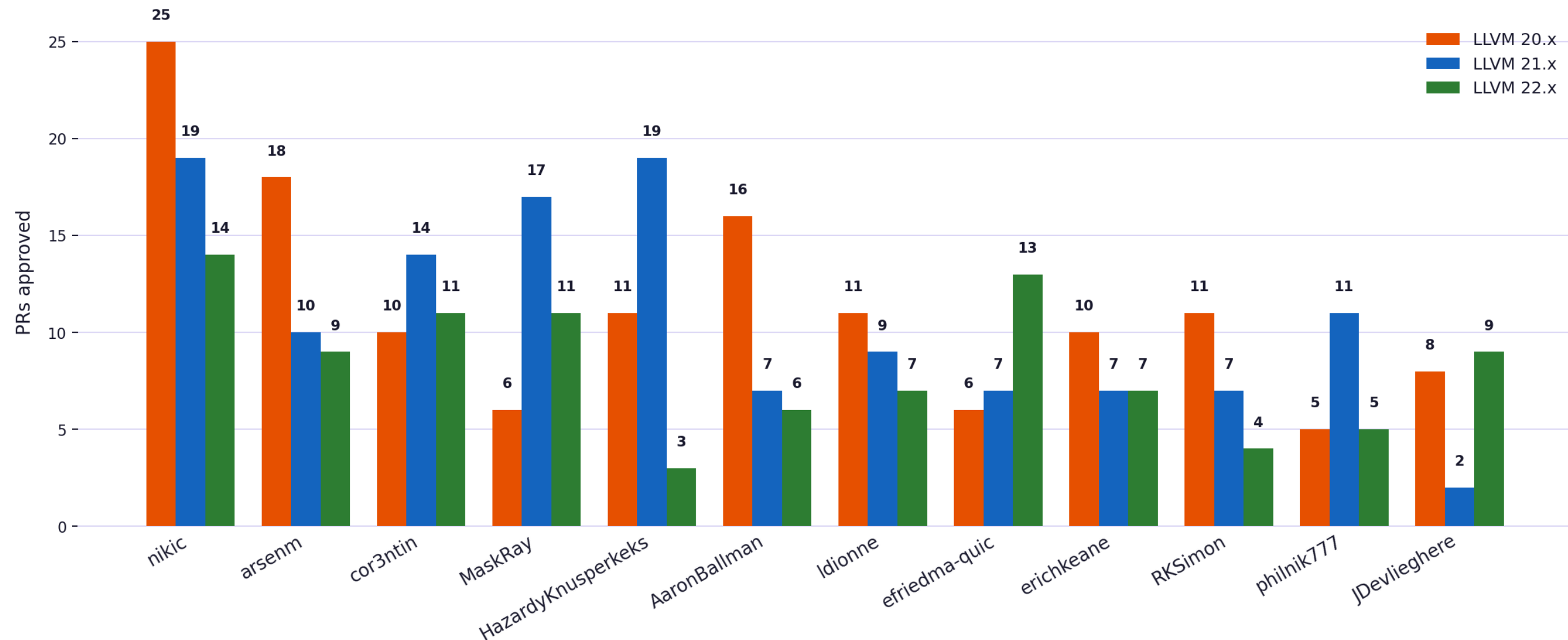
LLVM Core: Backends



LLVM Core: Subsystems

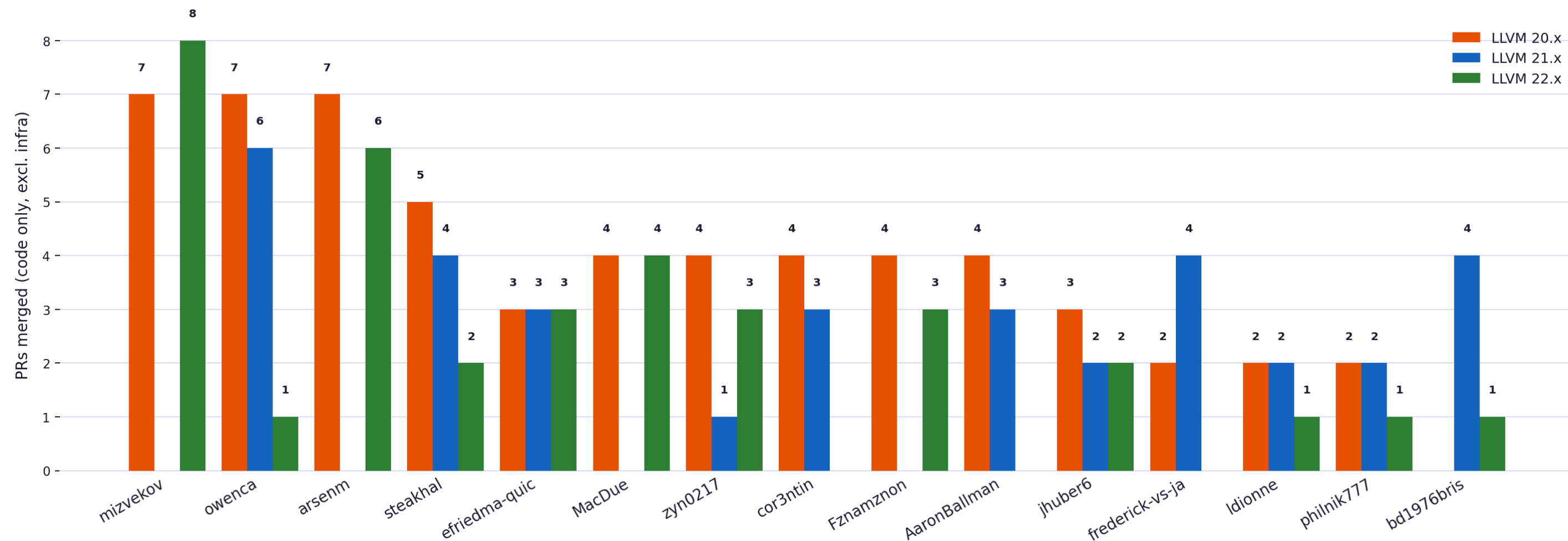


Who Reviews PRs



nikic alone: 58 PRs across 3 releases · top 5 reviewers handle the majority of all approvals

Who Submits PRs



~40 people do repeated work · most contributors appear only once

What Works

- Predictable cadence
- Community knows when to expect a release
- Milestone tracking on GitHub
- Release manager availability and responsiveness

What Doesn't Work

1. **No defined inclusion criteria** — every PR is a judgement call
2. **Flaky CI** blocks binary releases even when the code is ready
3. **Accidental ABI breakage** slips through reviews
4. **RC phase PR flood** — nobody looks until RC1 ships

Join us at the Roundtable

Right after this talk

What works? What should change? Come tell us.

Q&A
